

Mahindra University Hyderabad
École Centrale School of Engineering
Minor-II exam

Program: B. Tech. **Branch:** CSE, AI, ICS, CAM, ECE, VLSI, COM **Year:** II **Semester:** II
Subject: Theory of Computation (CS2204)

Date: 17-04-2026
Time Duration: 1.5 Hours

Start Time: 10:00 AM
Max. Marks: 40

Instructions:

- 1) All parts of a question should be answered consecutively.
- 2) Any question attempted using pencil will not be considered for the evaluation.
- 3) Use of any unfair means will result in severe disciplinary action.

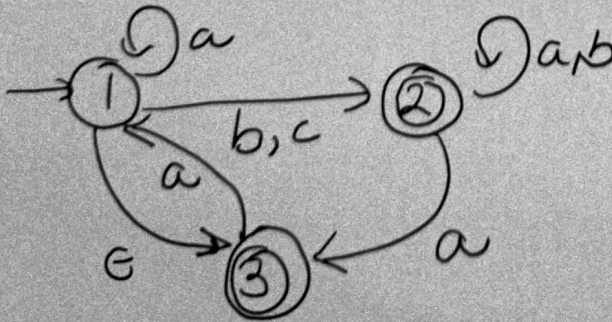
Q1.

[2*5=10M]

- a) Find a regular expression for the language over $\{0,1\}$
 $L = \{w \mid w \text{ contains at least one } 1 \text{ and at most two } 0\text{'s}\}.$
- b) Convert $(01)^* (10^* \cup 0)$ into an epsilon-NFA. We will not try to optimize the NFA or use our intuition about the language, but rather just use the constructions we saw in class for closure, to do the constructions. The order of operation is $()$, $*$, $\cup$, $\circ$. Hence, let's break down the regular expression appropriately.
 - 1) Build epsilon-NFA that recognizes $(01)^*$
 - a. First, construct epsilon-NFA that recognizes 01
 - b. Apply Kleene star to get a epsilon-NFA that recognizes $(01)^*$
 - 2) Build epsilon-NFA that recognizes $(10^* \cup 0)$
 - a. Build epsilon-NFA that recognizes (10^*)
 - b. Build epsilon-NFA that recognizes 0
 - c. Apply union operation to get epsilon-NFA that recognizes $(10^* \cup 0)$
 - 3) Apply concatenation to get epsilon-NFA that recognizes $(01)^* (10^* \cup 0)$

Q2. Convert the following three-state Automaton to Regular Expression using G-NFA method.
The order of selection of qrip (states to be eliminated) is $\{3,2,1\}$

[10M]



Q3.

[10M]

A Bluffer claims:

"All languages over the single-letter alphabet $\Sigma = \{a\}$ are regular".

He presents two similar-looking languages as evidence. While these two languages look similar, you know one of them is not regular and you want to bust him mathematically.

- $L_1 = \{a^{2n} \mid n \geq 0\}$ (The set of strings with an even number of a 's)
- $L_2 = \{a^{n^2} \mid n \geq 0\}$ (The set of strings where the number of a 's is a perfect square)

1. **Identifying Regularity (2 Marks)**

Which of these two languages is regular? Prove your choice by providing the formal 5-tuple description $(Q, \Sigma, \delta, q_0, F)$ of a Deterministic Finite Automaton (DFA) that recognizes it.

2. **Pumping Lemma Setup (3 Marks)**

For the other language in the scenario, you wish to use the Pumping Lemma to find a contradiction. Assume the language is regular and can be recognized using a DFA with p states.

- Suppose you select a string x from the language. What is the condition on the length $|x|$ required by the lemma?
- Suppose you decide to split the string into three pieces $x = uvw$. What are the required conditions on the lengths $|uv|$ and $|v|$?
- What is the condition that must be satisfied by the strings $uv^i w$ for all $i \geq 0$ if the language is regular?

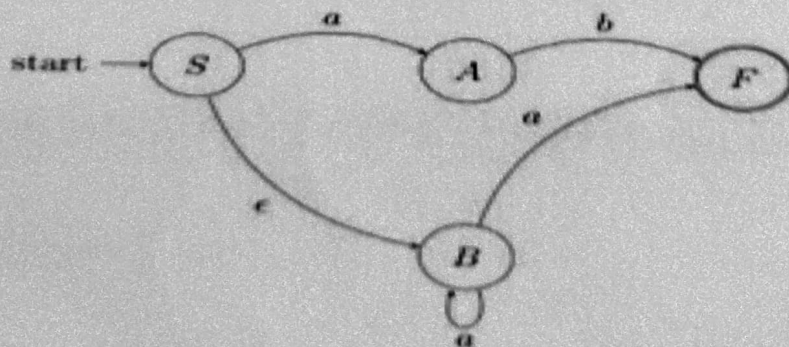
3. **Proving Non-regularity (5 Marks)**

Continue the proof by contradiction for the non-regular language.

- Select a specific string x from the language that satisfies the length requirement from Question 2(a). Explain why this specific choice is valid and useful for this proof.
- Suppose x is divided into uvw according to the constraints identified in Question 2(b). If we "pump up" to $x' = uv^2w$, prove that the length $|x'|$ no longer allows x' to be a member of that language.

Q4.

[2*5=10M]



- Generate the right linear grammar for the above stated finite automaton.
- Convert the generated right linear grammar in part (a) to the left linear grammar. Explain the entire conversion of right-linear grammar to left-linear grammar step by step.